Youfu Yan

New York, NY | arnoldyyf@gmail.com | linkedin.com/in/youfuyan | github.com/youfuyan | youfuyan.github.io

Summary

Software engineer building production-grade AI integrations, LLM evaluation pipelines, GPU inference prototypes, and reusable AI-agent workflows at Amazon. Experienced in the realities of productionizing AI: reliability under real traffic, deterministic verification, staged rollouts, security controls, and human release ownership. Built retrieval-augmented agentic systems in research (IEEE VIS 2024 Honorable Mention, equal-contribution first author). Strong backend and systems fundamentals; ships end-to-end with measurable outcomes.

Technical Skills

Production AI Systems: GPU inference and model serving, LLM evaluation and benchmark design, multi-model orchestration, agentic workflows and tool use, RAG and knowledge grounding, MLOps and AI observability, computer vision
AI/ML Stack: Python, PyTorch, AWS Bedrock, Amazon SageMaker, FastAPI, Neo4j
Backend & Cloud: Java, Spring Boot, REST APIs, AWS Lambda, DynamoDB, S3, SQS/SNS, event-driven architectures, AWS CDK, Docker, CloudWatch, CI/CD, multi-region deployment
Product & Quality: TypeScript, JavaScript, React, Redux, Next.js, React Native, Playwright, Cypress, SQL, load/stress/E2E testing

Experience

Software Development Engineer — June 2024 – Present
Amazon — *New York, NY*

AI/ML Migration & Evaluation | SageMaker, Bedrock, Python, PyTorch — 2026

- Led the evaluation and migration design for a high-volume background-removal path serving **hundreds of thousands of uncached images monthly**; proposed an in-account AWS architecture projected to reduce run-rate by **90%+** versus the third-party API.
- Built a **GPU SageMaker real-time endpoint** and a Python orchestration harness that dispatches to five model/service backends in parallel, comparing output quality, client latency, server-side model latency, and operating cost — recommending the migration candidate on evidence.
- Designed a blinded **LLM-as-a-Judge evaluation pipeline** over 500 production images using five quality criteria, three-vote majority, and pairwise position-swap controls; diagnosed an input-representation flaw responsible for approximately **70–79% false-failure verdicts**, fixing a reliability gap in the evaluation system.
- Operationalized reusable AI-native team workflows: contributed to an agent-driven build/deploy/browser-test loop with UI and CloudWatch assertions, and extended a production-support agent skill with identifier-only diagnosis, read-only production checks, evidence-based routing, human-approved writes, and **7 benchmark scenarios**; coordinated **5+ coding agents** to ship **6 reviewed changes in two days**.

Distributed Systems & Production Reliability | Java, AWS, DynamoDB, React — 2024–2026

- Owned a business-critical API launch in a new AWS region and executed a staged **1% to 100% traffic shift**; detected a live certificate mismatch at 20%, rolled back safely, coordinated remediation, and completed launch with **no customer-facing outage**.
- Designed and built the first version of an **async event-driven** creator-notification service used across three teams, including event-time DynamoDB conditional writes and retry/DLQ handling to preserve correctness under out-of-order events.

Software Development Engineer Intern — June 2023 – August 2023
Amazon — *New York, NY*

- Built a cross-platform real-time notification system with Java, React Native, AWS Lambda, and SQS, reducing content-status notification latency **60%** from 150 ms to 60 ms across iOS and Android.

Research Assistant – AI/ML & Human-Centered Computing Lab — September 2023 – May 2024
University of Minnesota, Twin Cities — *Minneapolis, MN*

- Co-developed KNOWNet, a **retrieval-augmented agentic system** for health information: the LLM agent reasons over a Neo4j Knowledge Graph to decide what to retrieve, validates answers against structured knowledge, and presents progressive D3.js visualizations; evaluated with a user study showing a **20% improvement in comprehension**.

Publication

KNOWNET: Guided Health Information Seeking from LLMs via Knowledge Graph Integration — 2024

- **Youfu Yan***, Yu Hou*, Yongkang Xiao, Rui Zhang, Qianwen Wang (*equal contribution) | *IEEE TVCG* | **IEEE VIS 2024 Honorable Mention**

Education

University of Minnesota, Twin Cities — Minneapolis, MN
M.S. Computer Science, GPA 3.95 | TA: Data Structures & Algorithms — *2024*

University of Minnesota, Twin Cities — Minneapolis, MN
M.A. Quantitative Methods in Education — *2022*

University of California, Santa Barbara — Santa Barbara, CA
M.A. Statistics — *2020*

Certification: AWS Certified Solutions Architect